

AI-powered prediction of friction and wear in functionalized Epoxy-MWCNT composites

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ABSTRACT

This study investigates the self-lubricating properties of epoxy composites reinforced with multi-walled carbon nanotubes (MWCNTs) functionalized with carboxyl (COOH), amine (NH₂), and silane coupling agents. Advanced artificial intelligence (AI) methods were employed to predict key tribological parameters coefficient of friction (COF) and wear rate as well as to classify wear mechanisms. Wear tests were conducted using a linear reciprocating tribometer under dry sliding conditions with a chromium steel ball as the counter surface. To simulate demanding conditions beyond typical industrial polymer bearings, sliding frequencies of 2, 5, and 8 Hz were applied under normal loads of 5, 10, and 15 N. The resulting data were analyzed using AI tools to identify patterns and predict tribological performance.

An Artificial Neural Network (ANN) was developed to model the relationships between COF, wear rate, experimental conditions, mechanical properties, and MWCNT composition, achieving high predictive accuracy ($R^2 = 0.98$ for COF; $R^2 = 0.78$ for wear rate). A Recurrent Neural Network (RNN) was used to capture the temporal evolution of COF under dominant wear mechanisms, including abrasion, adhesion, fatigue, and severe delamination. Additionally, a Convolutional Neural Network (CNN) accurately classified wear mechanisms from scanning electron microscopy (SEM) images. These AI-driven approaches provide a robust predictive framework for understanding and optimizing the tribological behavior of functionalized MWCNT-epoxy composites.

1. Introduction

Tribology, as described by Peter H. Jost in 1966, is the study of surface interactions that involve friction, wear, and lubrication [1]. This topic has now extended to include nanotribology, biotribology, and biomimetic tribology, demonstrating its interdisciplinary nature. Despite advances, tribology is still primarily empirical due to the complexities of friction and wear mechanisms. Modern computer tools have created new opportunities for data-driven analysis, resulting in novel ideas such as “Triboinformatics” and “Intelligent Tribology,” which seek to decipher complicated tribological phenomena using advanced data analytics [2,3].

Polymer tribology is distinct from metal and ceramic tribology due to the largely elastic character of polymer/metal or polymer/polymer interactions. Greenwood and Williamson (1976) [4] introduced the Plasticity Index (j) to forecast the first phase of plastic flow in such interactions.

$$j = \left(\frac{E}{H} \right) \sqrt{\left(\frac{S^*}{r} \right)}$$

Equation 1

Here,

- E : Young’s Modulus (a primary determinant of elastic behavior),
- H : Hardness,
- S^* : Standard deviation of asperity heights,
- r : Assumed spherical surface radius.

The ratio E/H separates polymers from metals and ceramics. Metals generally possess a E/H ratio above 100, whereas polymers exhibit a reduced E/H ratio of approximately 10, signifying their enhanced elastic deformation during contact. The friction coefficients (μ) for non-additive polymers in contact with other polymers, metals, or ceramics typically vary from 0.1 to 0.6, with PTFE exhibiting a minimum value of 0.1 [4].

The wear rate and coefficient of friction (COF) of materials are

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significantly influenced by their brittleness and toughness. Brittle materials with low toughness, such as epoxy matrices, have a limited ability to absorb and dissipate mechanical energy during sliding contact, making them prone to microcracking, fracture, and material removal. This leads to higher wear rates and rougher worn surfaces, which, along with wear debris, increase interfacial friction and elevate the COF [5,6]. For epoxy matrices, the COF typically ranges between 0.4 and 0.6 [7,8] due to their brittle nature and low toughness. Under high-speed sliding conditions, material removal occurs through various wear mechanisms, including abrasive wear, where hard asperities or particles scratch the surface, adhesive wear, caused by strong interfacial connections leading to material transfer, and fatigue wear. In contrast, tougher materials can deform plastically, absorbing energy without cracking, which helps maintain smoother contact surfaces and stable tribological behavior. Thus, the brittle nature and low toughness of materials like epoxies contribute to increased wear and higher friction during sliding due to poor resistance to crack initiation and propagation [9,10].

The COF characterizes the frictional behavior of epoxy composites by quantifying the friction force to the normal load. High COF values in unaltered epoxy materials cause frictional heat and energy losses. Adhesion and deformation are the main components to COF, with adhesion caused by molecular contacts at the interface and deformation caused by elastic and plastic asperity interactions. Surface roughness, load, and sliding speed all have a substantial effect on COF readings. Smooth surfaces generate more adhesion-related friction, whereas rougher surfaces encourage deformation-related friction [11]. Epoxy composites reinforced with functionalized Multiwall Carbon Nano Tubes (MWCNTs) have shown promise as self-lubricating systems [12, 13]. Functionalized MWCNTs, such as carboxyl (COOH), amine (NH₂) and silane-modified versions [14,15], improve load transmission while reducing friction and wear in the epoxy matrix. As mentioned in the literature, the tribological behavior of epoxy composites involves wear mechanisms, wear debris generation, and the formation of a tribo-film. During sliding, wear debris often containing fillers like MoS₂ can compact and form a protective tribo-film on the surface. This film reduces direct contact with the counterface, thereby lowering friction and wear in epoxy matrix [16].

Adhesion and deformation at the contact interface affect the COF and wear, which are important metrics in tribology. Functionalized MWCNTs improve tribological performance by minimizing adhesion and avoiding excessive deformation [17]. Abrasive wear, adhesive wear, and fatigue wear are the key wear mechanisms that occur depending on the loading and environmental circumstances [18].

These composites are rapidly being employed in aerospace and automotive applications due to their lightweight, high-strength, and structural integrity [19]. Artificial intelligence (AI) and machine learning have transformed tribology's predictive capabilities. Artificial Neural Network (ANN) models predict features include COF and wear rate using input variables such as material composition, hardness, and test circumstances [20]. Recurrent Neural Networks (RNNs), on the other hand, excel in detecting time-dependent fluctuations in tribological performance, such as the evolution of COF over time. Together, these models provide a powerful framework for understanding and optimizing the tribological behaviour of self-lubricating composites [21]. Meanwhile, Convolutional Neural Networks (CNNs) have transformed wear mechanism identification by analyzing microstructural images. In recent years, CNNs have emerged as a powerful tool for automated wear classification, offering high accuracy, feature extraction capabilities, and robustness in image-based tribological analysis [22–24].

Recent publications have significantly advanced the understanding of tribological behavior in abrasive belt grinding. One study developed an in-process tool condition monitoring system using support vector machines and genetic algorithms, enabling real-time detection of tool wear to enhance process efficiency and surface quality [25]. Another comprehensive review explored artificial intelligence techniques for

modeling and monitoring abrasive finishing processes, emphasizing machine learning's critical role in optimizing tribological performance and process control [26]. A separate study presented predictive modeling of material removal, analyzing how process parameters impact wear characteristics using statistical methods [27]. Additionally, a regression-based model for material removal was proposed, focusing on optimizing process parameters to achieve consistent tribological outcomes [28]. These findings highlight the importance of advanced modeling techniques in improving tribological performance and process efficiency.

In tribology, machine learning models are trained using input data such as material properties, operating conditions, and surface interactions to predict tribological outputs like wear and friction. Material composition, hardness, and microstructure influence wear behavior, while load, speed, and lubrication affect friction. Proper feature selection, high-quality data, and algorithm choice determine prediction accuracy, while overfitting and computational power impact model reliability. The epoxy resin's chemical structure significantly affects its tribological properties. The crosslinking density, a measure of the extent of chemical bonding between epoxy chains, directly influences the material's rigidity and resistance to deformation. Higher crosslinking densities generally lead to increased hardness and improved wear resistance [15]. The molecular weight of the epoxy resin also plays a role; higher molecular weight polymers often exhibit improved mechanical properties, translating to better tribological performance [13]. The type of hardener used in the curing process significantly impacts the epoxy matrix's final network structure and properties. Different hardeners can lead to variations in the glass transition temperature (T_g), the temperature at which the epoxy transitions from a glassy to a rubbery state [14]. Furthermore, the presence of specific functional groups within the epoxy structure can impact its interaction with the carbon reinforcement and the opposing surface during tribological contact, potentially influencing adhesion and friction [15]. The polarity of the epoxy resin, determined by the presence of polar functional groups, can also affect its interaction with the counterface material and any lubricant present, influencing friction and wear mechanisms.

The type of carbon reinforcement (e.g., carbon fibers, carbon nanotubes (CNTs), graphene) profoundly impacts the composite's tribological characteristics. Carbon fibers, known for their high strength-to-weight ratio, can significantly improve the wear resistance of epoxy composites. CNTs, with their high aspect ratio and excellent mechanical properties, offer significant potential for enhancing the tribological performance of epoxy composites. Their high surface area allows for strong interfacial bonding with the epoxy matrix, leading to improved load transfer and wear resistance [29]. Graphene, a single layer of carbon atoms arranged in a honeycomb lattice, also exhibits exceptional mechanical and thermal properties, making it a promising reinforcement for epoxy composites. However, the tendency of graphene sheets or CNTs to agglomerate can hinder their effectiveness in enhancing tribological performance [30]. The morphology of the reinforcement, such as fiber diameter, length, and aspect ratio, influences the interfacial bonding with the epoxy matrix and the overall mechanical properties, thereby affecting friction and wear.

The surface treatment of the carbon reinforcement significantly influences its interfacial interaction with the epoxy matrix and consequently, the composite's tribological behavior. Surface oxidation or functionalization can modify the surface chemistry of the reinforcement, enhancing its wettability and improving adhesion with the epoxy resin [31]. This improved adhesion can lead to better load transfer and reduced interfacial debonding during tribological contact, resulting in enhanced wear resistance. Conversely, inadequate surface treatment can lead to poor interfacial bonding, resulting in weak points within the composite and increased wear [32]. The choice of surface treatment technique should be tailored to the specific type of carbon reinforcement and the desired tribological properties.

The hardness and elastic modulus of the epoxy-carbon composite

directly influence its wear resistance. Higher hardness generally implies greater resistance to deformation and wear under load. Therefore, an optimal balance between hardness and elastic modulus is essential for achieving both high wear resistance and low friction [33]. The tensile strength and fracture toughness of the composite are crucial factors in determining its tribological performance. High tensile strength ensures that the material can withstand the stresses imposed during tribological contact without fracturing. High fracture toughness enhances the composite's resistance to crack propagation and catastrophic failure, leading to improved wear resistance [34]. The carbon reinforcement significantly improves these mechanical properties, reinforcing the matrix and preventing crack propagation. However, the distribution and orientation of the reinforcement play a crucial role in determining the overall tensile strength and fracture toughness of the composite. Uniform distribution and proper alignment of the reinforcement fibers can significantly enhance these properties, leading to improved tribological performance. The interfacial bonding between the epoxy matrix and the carbon reinforcement also plays a significant role, influencing the load transfer efficiency and the resistance to debonding and crack propagation [35].

The applied load and sliding speed during tribological testing are critical variables. Increasing the load increases the contact pressure, potentially leading to increased wear. Higher sliding speeds can generate more heat, potentially softening the epoxy matrix and accelerating wear. The combination of load and speed defines the tribological regime, influencing the wear mechanisms involved. At low loads and speeds, adhesive wear may be dominant, while at high loads and speeds, abrasive or fatigue wear may become more significant. Careful control of load and speed is essential to ensure that the tribological testing conditions are representative of the actual application environment. The selection of appropriate load and speed parameters should be based on the anticipated operating conditions of the composite material [36,37].

Tribology of epoxy composites is influenced by multiple factors, including material properties, operating conditions, surface interactions, and mechanical behavior. The crosslinking density, molecular weight, and functionalization of epoxy resins impact wear resistance and friction, while carbon reinforcements such as CNTs, graphene, and fibers enhance performance but require proper dispersion and interfacial bonding. Mechanical properties like hardness, elastic modulus, and fracture toughness play a crucial role in determining tribological behavior. Given the increasing demand for self-lubricating epoxy composites with superior wear resistance, integrating AI-driven predictive models and automated wear classification presents a transformative approach in tribology.

While prior studies have explored individual AI techniques for wear prediction [38–40], a comprehensive framework incorporating ANNs for predictive modeling, RNNs for wear evolution analysis, and CNNs for Scanning Electron Microscope (SEM) based wear classification has not been fully developed. This study addresses this gap by employing an AI-enhanced predictive and classification framework for epoxy composites reinforced with COOH, NH₂, and silane-functionalized MWCNTs. The proposed approach utilizes ANNs to predict tribological performance, RNNs to model time-dependent COF behavior, and CNNs to classify wear mechanisms with high accuracy. By integrating these advanced AI methodologies, this work enhances automation and reliability in tribological assessments, providing valuable insights for optimizing self-lubricating nanocomposites. Ultimately, this study demonstrates the potential of AI to improve predictive accuracy, wear mechanism recognition, and material optimization, paving the way for next-generation tribological systems in aerospace, marine, and high-performance engineering applications.

2. Experimental section

2.1. Materials and sample preparation

Bisphenol A/epichlorohydrin 50–70 wt% epoxy resin, Isophoronediamine (3-aminomethyl-3,5,5-trimethyl-cyclohexylamine) 50–60 wt% curing agent, and the mold release agent were purchased from Norski, Plimmerton, New Zealand. Silane coupling agent KH-550 (C₂H₅O)₃Si₃H₆NH₂ and H₂SO₄ and HNO₃ - 63w% Concentration, Ethanol (CH₃CH₂OH) and acetone (CH₃)₂CO with 99.9 % purity was obtained from Merck Life Science Ltd., Auckland, New Zealand. Pristine MWCNTs and NH₂ functionalized MWCNTs had ~10 μm in length and diameter less than 40 nm, and purity >98 % was obtained from Adnano Technologies Pvt.Ltd., Karnataka, India.

2.2. Sample preparation and characterization

Pristine MWCNTs (30 mg) were functionalized by treatment with a 3:1 sulfuric-nitric acid mixture, stirred magnetically at 40 °C for 10 h. The product was filtered, washed repeatedly with deionized water, and purified via centrifugation at 1000 rpm for 7 min until the supernatant reached pH 7.0. The resulting COOH-MWCNTs were dried under vacuum at 90 °C for 15 h to remove moisture and solvents. A silane-ethanol solution (0.001 g/mL) containing 10 wt% KH-550 relative to COOH-MWCNTs was prepared. Dried COOH-MWCNTs (15 mg) were dispersed in this solution, stirred at 60 °C for 30 min, and sonicated for 1 h to ensure uniformity, forming Si-O-C bonds. Excess solvent was removed by rinsing with acetone, followed by vacuum drying at 70 °C for 8 h and an additional 15 h at 90 °C.

Epoxy composites were fabricated via open molding, incorporating P-MWCNTs, COOH-MWCNTs, NH₂-MWCNTs, and silane-modified MWCNTs at 0.1, 0.3 and 0.5 wt%. MWCNTs were dispersed in epoxy resin by mechanical stirring (1 h) and ultrasonication (20 min) using a 550 W, 20 kHz horn transducer. Isophoronediamine curing agent was added (4:1 epoxy: hardener), followed by degassing at –1 bar of vacuum pressure for 10 min. The mixture was cast into aluminum molds treated with a release agent, cured at 10 °C overnight, and post-cured at 60 °C for 12 h to ensure crosslinking [41]. Control samples without MWCNTs were prepared for comparison.

2.3. Tensile testing

Dog-bone-shaped tensile samples were prepared following the EN ISO 527-2 type 1b standard [42]. Each specimen measured 150 mm in length, 4 mm in width, with a gauge length of 50 mm. A computer-controlled servo-hydraulic universal testing machine, capable of a 50 kN maximum load, was used for tensile tests at a crosshead speed of 1.5 mm/min. Stress-strain data were collected to evaluate the ultimate tensile strength, Young's modulus, and elongation of epoxy composites, adhering to ISO 527-1 [43] guidelines. Hardness measurements were conducted using a type D durometer per ASTM D2240, with average values derived from five measurements per condition [44].

2.4. Tribological testing

Rectangular plates, sized 5 mm in height, 40 mm in length, and 25 mm in width, were fabricated for tribological testing. Dry sliding wear tests were performed using an Rtec MFT-5000 tribometer (Rtec Instruments, USA). A calibrated loading mechanism ensured consistent normal force application during linear reciprocating tests. The tribometer's motor controller precisely adjusted sliding frequencies, while a built-in load cell continuously recorded friction forces for real-time data. The counterface consisted of a 10 mm diameter chromium steel ball (AISI 52100) with a surface average roughness (R_a) of 2.5 μm determined using Talysurf 50 surface profilometer. The hardness of the ball was approximately 60 HRC, consistent with standard specifications for

AISI 52100 bearing steel [45]. Tests were conducted at 24 °C and 60 % relative humidity, with composite surfaces and counter-balls cleaned using acetone to remove contaminants.

Test conditions were designed to exceed typical industrial polymer bearing requirements [46,47]. Sliding frequencies of 2 Hz, 5 Hz, and 8 Hz corresponded to sliding speeds of 0.02 m/s, 0.07 m/s, and 0.2 m/s, respectively. Normal loads of 5 N, 10 N, and 15 N were applied, resulting in maximum contact pressures of 121.73 MPa, 153.39 MPa, and 175.58 MPa, calculated via Hertzian contact theory [48]. Test durations of 68, 95, and 248 min ensured a consistent sliding distance of 400 m. Before sliding wear tests, all samples were polished using 1200 grit paper to ensure a consistent surface finish with a surface average roughness (R_a) of 6.5 μm . Each test condition was repeated three times for reliability, and the COF was determined with standard errors. Post-test analysis involved scanning electron microscopy (SEM) using a Hitachi SU-70, with samples platinum-coated for 60 s using a Hitachi E-1045 ion sputter. SEM imaging was performed at a 10 kV accelerating voltage.

3. Methodology

3.1. Artificial neural network

This study uses a feedforward ANN to predict COF and Wear Rate of MWCNT-reinforced epoxy composites under varied mechanical and tribological conditions. ANNs excel at modeling complex, non-linear relationships in friction, wear, and lubrication where traditional models falter due to oversimplified assumptions.

The dataset includes 65 samples with six inputs MWCNT wt%, sliding speed (ms^{-1}), load (N), tensile strength (MPa), compressive strength (MPa), hardness (shore D) and outputs COF and Wear Rate (mm^3/m). Inputs and targets are normalized to $[-1, 1]$ as per equation (2), via MATLAB's mapminmax to ensure scale equity and training stability.

$$X_{\text{norm}} = \frac{2(X - X_{\min})}{X_{\max} - X_{\min}} - 1 \quad \text{Equation 2}$$

where.

X - Original value of the input or output variable
 $X_{\min}$, $X_{\max}$ - Minimum and maximum values of the variable
 X_{norm} - Normalized value

The ANN, a multilayer architecture, has three hidden layers (20, 15, 8 neurons) with tansig functions for non-linearity and a linear output layer for regression as shown in equation (3). The Levenberg-Marquardt (LM) algorithm [49] is a second-order optimization method that minimizes Mean Squared Error (MSE) by combining gradient descent and the Gauss-Newton method for improved convergence stability (Haykin, 2009) [27]. The ANN is trained for a maximum of 100 epochs, with an early stopping criterion if the MSE drops below 10^{-5} or the gradient falls below 10^{-7} , ensuring efficient learning. The 10-epoch validation check prevents overfitting by stopping training if the validation error does not improve for 10 consecutive epochs. To ensure generalization, the dataset is split into 70 % for training, 15 % for validation, and 15 % for testing, enabling robust model evaluation.

ANN activation function

$$\text{tansig}(z) = \frac{2}{1 + e^{-2z}} - 1 \quad \text{Equation 3}$$

where.

$z = \sum w_i x_i + b$: Weighted sum of inputs
 w_i : Weights
 x_i : Inputs
 b : Bias

Post-training, denormalized predictions are assessed via MSE (4), RMSE (5), and R^2 (6) for COF and Wear Rate, capturing accuracy for frictional resistance and material loss.

$$(MSE) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad \text{Equation 4}$$

where.

n : Number of samples
 y_i : Actual output
 $\hat{y}_i$: Predicted output

Root Mean Squared Error (RMSE)

$$(RMSE) = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad \text{Equation 5}$$

Coefficient of Determination (R^2)

$$(R^2) = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{Y})^2} \quad \text{Equation 6}$$

where.

$\bar{Y}$: Mean of actual output

Sensitivity analysis evaluates the influence of input variables on model outputs by systematically perturbing each input by 10 % while keeping others constant. The resulting changes in the output are normalized and plotted to facilitate comparison across different input parameters. This process helps to identify the most dominant factors, where inputs that cause the largest variations in the output are deemed the most influential. By analyzing these effects, researchers can gain deeper insights into the model's behavior, refine input selection, and improve the overall interpretability and reliability of the predictive framework. This approach merges empirical data with computational modeling, enhancing the understanding of friction and wear in MWCNT-epoxy systems for engineering applications.

3.2. Recurrence neural network

This study employed a RNN framework with Long Short-Term Memory (LSTM) layers in MATLAB to perform one-step-ahead prediction of the COF time-series data over a 6000-s interval. The COF was treated as a univariate sequence, with each prediction at time t based on the value at $t-1$. The process followed a standard time-series machine learning pipeline consisting of data acquisition, preprocessing, splitting, hyperparameter optimization, model training, prediction, evaluation, and visualization.

The COF dataset was imported from an Excel file containing Time and COF columns. The script validated the presence of these variables, extracted them as vectors, and standardized the time axis to 6000 s via linear interpolation when required. Raw COF values were retained without smoothing to preserve natural fluctuations. The COF values were normalized using z-score transformation to improve training stability and convergence [50]. Input-output pairs were generated for sequence-to-sequence regression: the input (X) comprised normalized COF values from index 1 to $N-1$, while the output (Y) contained the corresponding values from index 2 to N .

The COF values are normalized using z-score transformation to standardize the data to a mean of 0 and a standard deviation of 1. equation (7) for z-score normalization is:

$$Z_i = \left(\frac{X_i - \mu}{\sigma} \right) \quad \text{Equation 7}$$

where,

Z_i is Normalized COF value at index i .
 X_i is Original COF value at index i .
 μ is Mean of the COF values.
 σ is Standard deviation of the COF values.

The input-output pairs are created for one-step-ahead prediction, where the input (X) consists of normalized COF values from index 1 to $N-1$, and the output (Y) consists of normalized COF values from index 2 to N . For a time series of normalized COF values $Z = [Z_1, Z_2, \dots, Z_{(N-1)}]$, the input and output sequences are defined as equation (8):

$$X = [Z_1, Z_2, \dots, Z_{(N-1)}] \quad \text{Equation 8}$$

$$Y = [Z_2, Z_3, \dots, Z_{(N)}]$$

where,

- X : Input sequence of normalized COF values for time steps $t = 1$ to $N-1$
- Y : Target sequence of normalized COF values for time steps $t = 2$ to N .
- N : Total number of COF data points.

These equations capture the preprocessing steps described: z-score normalization ensures training stability, and the input-output pairing sets up the sequence-to-sequence regression task for one-step-ahead COF prediction.

Data was split chronologically to preserve temporal dependencies. 80 % of the series was allocated to the training set, with 20 % of this portion used for validation. The remaining 20 % of the series served as a test set for performance reference. No random shuffling was applied to prevent data leakage.

Hyperparameter Optimization was conducted over hidden units [100, 150, 200] and learning rates [0.005, 0.001, 0.0005], using validation RMSE as the selection criterion. For each configuration, the network architecture comprised:

Sequence input layer (univariate time series): Takes univariate time series $X_t = [X_t - k, \dots, X_t]$, where k is the sequence length fed to the BiLSTM (in context, often the entire sequence up to t).

Bidirectional LSTM layer (capturing forward and backward temporal context) with $h \in [100, 150, 200]$ hidden units process X_t forward ($h \rightarrow t$) and backward ($h \leftarrow t$), combining outputs as shown in equations (9) and (10).

For each time step t , the forward and backward LSTM outputs are:

$$t \rightarrow h = \text{LSTM}_f(x_t, \rightarrow h_{t-1}, \rightarrow c_{t-1}) \quad \text{Equation 9}$$

$$t \leftarrow h = \text{LSTM}_b(x_t, \leftarrow h_{t+1}, \leftarrow c_{t+1}) \quad \text{Equation 10}$$

where,

$h \rightarrow t$ Forward LSTM hidden state, processing the sequence from start to t .
 $h \leftarrow t$ Backward LSTM hidden state, processing the sequence from end to t .

These $h \rightarrow t$ and $h \leftarrow t$ follow the LSTM equations [51]. The hidden state ht in an LSTM is computed as equation (11):

$$ht = Ot \cdot \tanh(Ct) \quad \text{Equation 11}$$

where,

Forget gate: $ft = \sigma(Wf \cdot [ht - 1, xt] + bf)$
Input gate: $it = \sigma(Wi \cdot [ht - 1, xt] + bi)$
Cell state candidate: $C \sim t = \tanh(Wc \cdot [ht - 1, xt] + bc)$
Output gate: $ot = \sigma(Wo \cdot [ht - 1, xt] + bo)$
 xt : Input at time t
 $ht-1$: Previous hidden state
 ft, it, ot : Forget, input, and output gates
 $C \sim t$: Candidate cell state
 ht : Current hidden state
 Wf, Wi, Wc, Wo : Weight matrices
 bf, bi, bc, bo : Bias vectors
 σ : Sigmoid activation function
 $\tanh$: Hyperbolic tangent activation function

These bidirectional LSTM layers are for both long and short-term pattern capture. The optimal configuration was retrained with extended settings: 600 epochs, piecewise learning rate reduction (factor 0.2 every 150 epochs), tighter gradient threshold (0.5), and validation checks every 20 iterations. L2 regularization and dropout were maintained to enhance generalization. The final model was applied to the entire dataset. Predictions were denormalized, and performance was assessed using: RMSE, R^2 [52].

This approach ensured the model captured complex temporal behaviors such as cycles, trends, and abrupt changes commonly present in experimental COF measurements, while minimizing overfitting and maintaining realistic predictive performance. It provides a valuable tool for forecasting tribological behavior in self-lubricating epoxy-MWCNT composites, advancing material performance optimization.

3.3. Convolution neural network

This study employs a CNN adapted from Squeeze Net to classify four wear mechanisms abrasive, adhesive, fatigue, and material pull-outs in SEM images of self-lubricating epoxy-MWCNT composites. The dataset comprises 100 images, with 25 images per class. The dataset for each class was divided into 20 training images and 5 validation images, resulting in a total of 80 training images and 20 validation images across all classes. This corresponds to a training fraction of 20/25 (80 %) for each class, ensuring a balanced and representative split for CNN training and performance evaluation [53].

Training images are augmented as shown in equation (12) with random rotations (-30° – 30°), translations (-10 to 10 pixels), scaling (0.8 – 1.2), and shearing (-10° – 10°) via an image Data Augmenter to enhance model robustness [54].

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad \text{Equation 12}$$

where,

(x, y) : Original pixel coordinates
 (x', y') : Transformed coordinates
 a, b, c, d : Parameter for rotation $[-30^\circ, 30^\circ]$, scaling $[0.8, 1.2]$ and shearing $[-10^\circ, 10^\circ]$
 t_x, t_y : Translation $[-10, 10]$ pixels

Since Squeeze Net requires RGB inputs, grayscale SEM images are converted to RGB by triplicating the single channel, resized to 227×227 pixels, and stored in an augmented Image Datastore. Test images are similarly processed without augmentation. Squeeze Net's architecture is modified for transfer learning by replacing its final convolutional layer with a new convolution 2nd Layer for 4 classes, followed by a global Average Pooling 2nd Layer, softmax Layer, and classification Layer.

The CNN extract features via convolution.

$$S(i,j) = \sum_m \sum_n I(i+m,j+n) K(m,n) \quad \text{Equation 13}$$

where.

I : Input image
 K : Convolution kernel
 $S(i,j)$: Feature map value (i,j)
 m,n : Kernel dimensions

Training uses the SGDM optimizer with a 0.0001 initial learning rate, 0.9 momentum, and a piecewise schedule (dropping by 0.1 every 20 epochs) over 50 epochs. Validation occurs every 8 epochs using the test set, with early stopping after 10 unchanged validations to prevent overfitting. Post-training, the model predicts wear types for the test set, achieving high accuracy. To further validate performance, 24 images are randomly selected from the full dataset of 100 using random (100, 24), representing a mix of training and test samples. The CNN classifies these 24 images with 100 % accuracy, correctly identifying each wear mechanism (abrasive, adhesive, fatigue, or material pull outs) against their true labels.

Performance is evaluated with a confusion matrix for all 4 classes, plus per-class precision, recall, and F1-scores derived from the confusion matrix [55]. Overall accuracy, macro-averaged metrics, and visualizations, including a confusion chart and prediction plots for the 24 random images with true labels, predicted labels, and confidence scores, are generated. These results, grounded in deep learning principles,

demonstrate CNN's ability to accurately classify wear mechanisms in epoxy-MWCNT composites, with the random 24-image sample underscoring its predictive reliability.

The results of the CNN were evaluated by the loss function and the Accuracy, as shown in equations (14) and (15).

Loss function.

$$L = - \sum_{i=1}^N \sum_{k=1}^4 y_{i,k} \log(p_{i,k}) \quad \text{Equation 14}$$

where.

N : Number of images
 $Y_{i,k}$: True label
 $P_{i,k}$: Predicted probability for class k

Accuracy

$$\text{Accuracy} = \frac{\sum_{k=1}^4 C_{k,k}}{\sum_{i=1}^4 \sum_{j=1}^4 C_{i,j}} \quad \text{Equation 15}$$

$C_{k,k}$: Diagonal elements of the confusion matrix, representing correct predictions for class k .

$C_{i,j}$: General elements of the confusion matrix, representing instances where true class is i and predicted class is j .

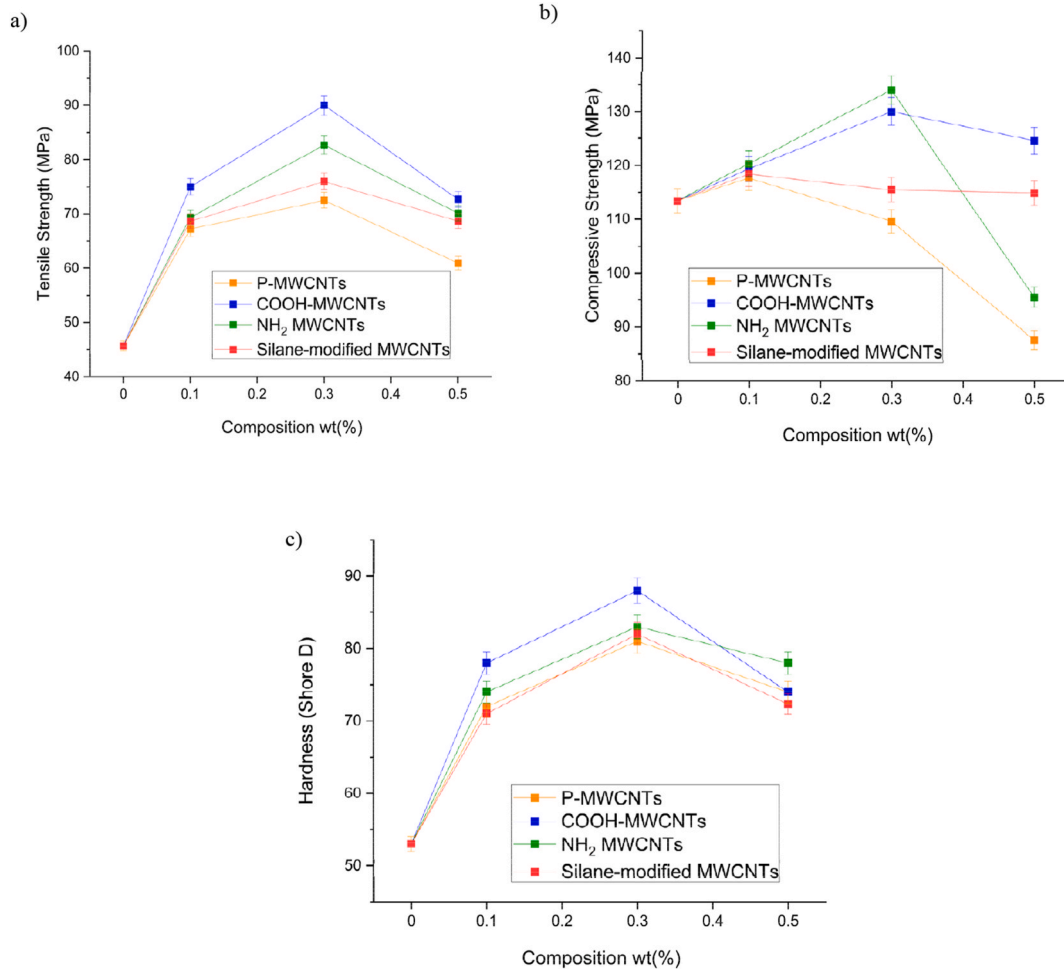


Fig. 1. (a) Tensile strength vs. composition and (b) Compressive Strength vs. composition (c) Hardness vs. composition of epoxy-MWCNT composites functionalized with COOH, NH₂, and silane modification. Error bars represent the standard error of the mean (n = 3).

4. Results and discussion

4.1. Mechanical properties

Fig. 1 explains the mechanical properties of epoxy composites reinforced with different MWCNT derivatives at varying compositions (0.1, 0.3, and 0.5 wt%) exhibit distinct trends across tensile strength, compressive strength, and hardness. In Fig. 1a, the highest tensile strength (68 MPa) is observed at 0.3 wt% COOH-MWCNT, significantly surpassing neat epoxy (45 MPa). This suggests enhanced load transfer efficiency due to strong interfacial bonding. Even at the 0.1 wt%, COOH-MWCNTs exhibit high tensile strength (60 MPa), outperforming neat epoxy. At 0.5 wt%, tensile strength decreases for all samples, likely due to particle agglomeration reducing stress distribution efficiency [41].

In Fig. 1b the maximum compressive strength (134 MPa) is achieved at 0.3 wt% with NH_2 -functionalized MWCNTs, highlighting their reinforcing efficiency under compression. COOH-MWCNTs also show significant improvement at 0.1 wt% (119 MPa) and 0.5 wt% (124 MPa), indicating stable compressive properties.

In Fig. 1c, the peak hardness value (88 Shore D) is obtained at 0.3 wt% with COOH-MWCNT reinforcement, demonstrating enhanced resistance to deformation. Even at 0.1 wt%, COOH-MWCNT (78 Shore D) provides a notable improvement over neat epoxy (53 Shore D). All compositions showed an improvement in hardness, with a significant increase observed across all compositions. Optimal Reinforcement at 0.3 wt% was observed across all mechanical properties; 0.3 wt% appears to be the most effective concentration for achieving maximum strength and hardness. COOH-MWCNT dominance consistently outperforms others in tensile and hardness improvements, likely due to better dispersion and epoxy adhesion.

4.2. Tribological properties

Fig. 2 illustrates the wear rate and COF as functions of MWCNT composition (wt%), comparing neat epoxy, non-functionalized MWCNTs, and functionalized MWCNTs (COOH, NH_2 , and Silane-modified) under 10 N load, 5 Hz frequency, and a sliding distance of 400 m. These results highlight the influence of MWCNT content and surface functionalization on epoxy composites' self-lubricating and wear-resistant properties.

Fig. 2a presents the COF variation with respect to composition (wt%), reflecting the frictional response of each composite. Neat epoxy exhibits the highest COF (~ 0.6), demonstrating its high adhesive wear tendency and poor lubrication ability. Upon the addition of MWCNTs, the COF initially decreases, reaching a minimum at ~ 0.3 wt%, followed by a slight increase beyond this concentration.

Among the functionalized MWCNT composites, COOH-MWCNTs exhibit the lowest COF (0.38 at 0.3 wt% composition), maintaining stability across different compositions. This result suggests that COOH-functionalization facilitates effective tribofilm formation [16,41], reducing direct asperity contact and enhancing lubrication. NH_2 -MWCNTs also show relatively low COF (0.45) but exhibit slight variations, likely due to localized adhesion and intermittent surface interactions during sliding. Silane-modified and non-functionalized MWCNT composites display higher COF values, indicating that their tribological behavior is less effective in reducing surface friction, possibly due to weaker interactions at the matrix-reinforcement interface.

Fig. 2b shows the wear rate versus composition, indicating a significant reduction in wear rate with the addition of MWCNTs, the neat epoxy exhibits the highest wear rate. This result suggests that, in the absence of reinforcing fillers, epoxy lacks sufficient resistance to material removal, leading to rapid surface degradation during sliding contact. With the incorporation of MWCNTs, wear resistance improves, particularly up to ~ 0.3 wt%, where all MWCNT based composites exhibit the lowest wear rate. Among the different MWCNT modifications, COOH-

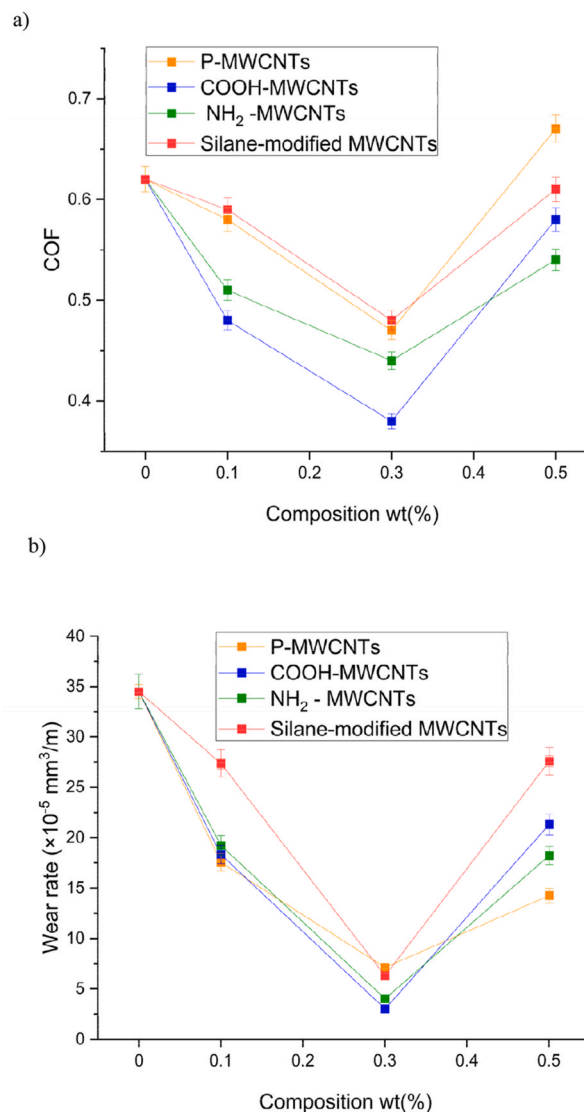


Fig. 2. (a) COF vs. composition and (b) Wear rate vs. composition of epoxy-MWCNT composites functionalized with COOH, NH_2 , and silane modification, tested at 10 N load, 5 Hz frequency, and a sliding distance of 400 m. Error bars represent the standard error of the mean ($n = 3$).

functionalized MWCNTs and NH_2 -functionalized MWCNTs exhibit superior wear resistance. This can be attributed to their strong interfacial bonding with the epoxy matrix, which enhances load transfer and prevents particle pull-out during wear. Conversely, non-functionalized and silane-modified MWCNTs show an increase in wear rate beyond 0.3 wt%, suggesting that MWCNT agglomeration at higher concentrations leads to non-uniform stress distribution, weakening the composite structure. These results indicate that optimal MWCNT dispersion and matrix reinforcement are achieved at ~ 0.3 wt% composition, beyond which wear performance deteriorates due to reduced interfacial efficiency.

A comparison between wear rate and COF trends highlights 0.3 wt% as the optimal composition, where both wear resistance and friction reduction are maximized. Beyond this threshold, increased MWCNT loading leads to agglomeration, which disrupts load transfer efficiency and causes frictional instability. Among all tested composites, COOH-functionalized MWCNTs exhibit the best balance between wear resistance and low COF, followed by NH_2 -MWCNTs. In contrast, silane-modified and non-functionalized MWCNTs show moderate improvement over neat epoxy but are more prone to performance deterioration

at higher concentrations.

These findings confirm that MWCNT functionalization plays a crucial role in determining the tribological performance of epoxy composites. The COOH-functionalized MWCNTs offer the most stable and effective reinforcement, making them ideal for self-lubricating applications where wear resistance and friction control are critical. This study demonstrates the importance of optimized nanofiller composition and surface modification strategies, providing valuable insights for designing high-performance epoxy-based tribological materials.

4.3. Artificial Neural Network

Fig. 3 presents the regression analysis of an ANN across four stages: Fig. 3a Training ($R = 0.99945$), Fig. 3b Validation ($R = 0.97894$), Fig. 3c Testing ($R = 0.72464$), and Fig. 3d Overall Performance ($R = 0.94464$). These scatter plots compare predicted outputs (Y-axis) against target values (X-axis), with the correlation coefficient (R) reflecting prediction accuracy. The Training phase exhibits an excellent fit ($R = 0.99945$), with points tightly clustered along the $Y = T$ line, indicating strong learning of the training data. Validation performance is slightly lower ($R = 0.97894$), still showing good generalization to unseen data, though with increased scatter. The Testing phase ($R = 0.72464$) reveals a significant drop, suggesting challenges in predicting new data, possibly due to overfitting or data variability. The overall performance ($R = 0.94464$) remains strong, although it has been influenced by the testing phase.

Fig. 4a compares predicted versus actual COF and Fig. 4b compares predicted versus actual Wear Rate values, with additional metrics

provided. For COF, the plot spans 0.35 to 0.65, with a Mean Squared Error (MSE) of 0.000276, Root Mean Squared Error (RMSE) of 0.016603, and Correlation R^2 of 0.982385, indicating highly accurate predictions as points align closely along the $Y = T$ line. The Wear Rate plot, ranging from 0.1 to 0.7, has an MSE of 0.008561, RMSE of 0.092527, and R^2 of 0.780183, showing reasonable accuracy but more scatter than COF, reflecting less precision. These metrics align with Fig. 3's trends, confirming the ANN's capability to model tribological behaviors, particularly for COF, with high accuracy across training and validation stages.

Fig. 5 compares ANN actual versus predicted values for COF and Wear Rate across 65 samples. Fig. 5a shows COF (0.35–0.7), with predicted values closely matching actual values, indicating strong performance despite minor deviations. Fig. 5b displays the Wear Rate (0.1–0.7 mm^3/m), where predictions follow the trend but show larger discrepancies, especially at peaks (samples 25, 40, 55), suggesting overestimation. The ANN predicts COF more accurately than Wear Rate, likely due to greater variability in the latter. This aligns with prior regression (COF $R^2 = 0.982385$, Wear Rate $R^2 = 0.780183$) and sensitivity analyses, highlighting the model's strengths and limitations in capturing tribological behaviors.

Fig. 6 explains the sensitivity analysis in the context of ANN and quantifies how much each input parameter influences the output of COF or Wear Rate. The normalized sensitivity (ranging from 0 to 1) reflects the relative change in the output when an input is varied, normalized by the maximum sensitivity observed across all inputs. Fig. 6a Sensitivity Analysis for COF and hardness (1.0) is the most influential, followed by

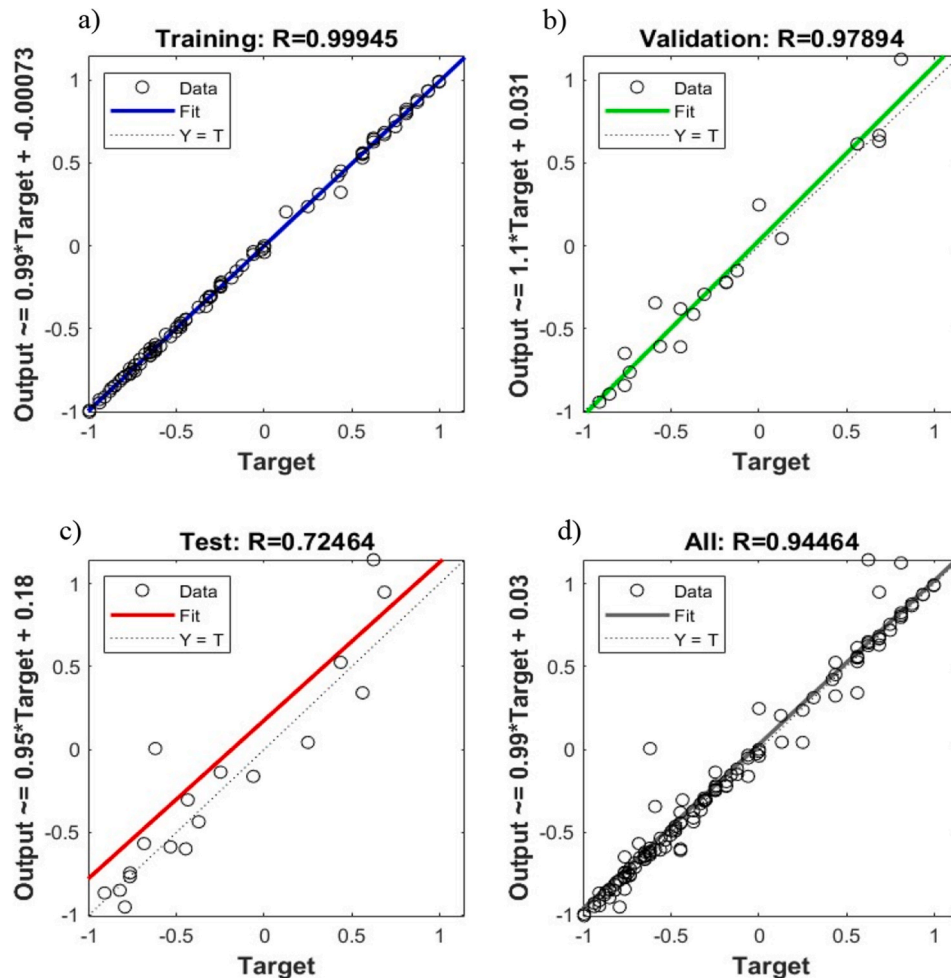


Fig. 3. Regression Analysis of ANN actual vs Predicted results a) for training, b) Validation, for c) Test and d) all the data.

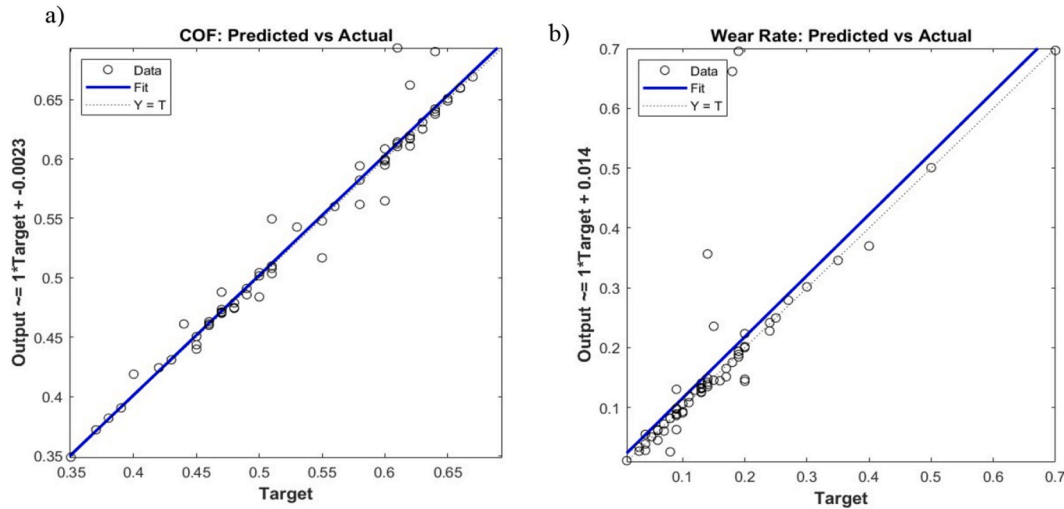


Fig. 4. Regression plot of a) COF and b) Wear rate.

load and compressive strength (0.7), tensile strength (0.5), and composition and speed (0.2), aligning with hardness governing friction. Fig. 6b For wear rate, hardness (1.0) leads, with load (0.8), compressive strength (0.6), tensile strength (0.5), composition (wt) % (0.2), and speed (0.1), reflecting hardness and load driving wear resistance.

4.4. Recurrent neural network results

The RNN with Bidirectional LSTM layers were employed to predict the time-dependent evolution of the COF for epoxy composites reinforced with functionalized MWCNTs. The tribological tests were conducted under a 10N applied load, 5 Hz sliding frequency, and 400m sliding distance, and were utilized as a comparative framework for evaluating different functionalization strategies. Figs. 7–11 illustrate the predicted and actual COF variations over the 6000-s test duration for MWCNT, COOH-MWCNT, NH₂-MWCNT, and silane-modified MWNT composites and neat epoxy. The COF vs time graphs represent experimentally measured COF values corresponding to RNN-predicted COF trends. The RNN model's accuracy in capturing COF fluctuations, stabilization trends, and peak values were evaluated by comparing predicted and actual data, with the RMSE quantifying the average prediction error.

Fig. 7 COF Prediction for pristine MWCNT composite displays a higher fluctuation. The COF fluctuates between 0.35 and 0.7, showing an initial peak, followed by oscillations, and a gradual decline. The predicted COF closely follows the actual trend, capturing major peaks and troughs, though with slight deviations, but the overall model suffers to closely follow the oscillating trends. The accompanying SEM image highlights abrasive wear, suggesting that the high COF values correlate with abrasive wear mechanisms, where surface material is removed due to friction, producing debris and tribo film. The cross-sectional view of the SEM image shows the wear depth and how the pits and grooves were carved into the material. The regression plot of the predicted and actual COF variations shows how accurately RNN captures the COF variation over time. Test RMSE = 0.0064: The Root Mean Squared Error between predicted and actual values for the unseen test set is 0.0064 (in the normalized scale). This indicates a low average prediction error. Test R² = 0.9713: The model explains ~97.13 % of the variance in the test data, meaning strong predictive performance on unseen data.

Fig. 8, corresponding to the COOH-MWCNTs epoxy matrix, compares actual and predicted COF over 6000 s, but with a different trend: COF starts around 0.45, drops to 0.38, and stabilizes around 0.37. This corresponds to the stable tribofilm formation [41], and then the COF drops down due to the self-lubrication. The predicted COF aligns well

with the actual values, showing the RNN's ability to capture this distinct behavior. The SEM top-view image reveals slight fatigue wear, evidenced by crack lines-oriented perpendicular to the wear surface. In contrast, the cross-sectional view shows a stable protective film of approximately 20 μm thickness on top of the base material, with no visible surface cracks penetrating into the subsurface. This indicates that the COOH-functionalized MWCNTs provided strong reinforcement and promoted a self-lubricating mechanism, reducing crack propagation and material removal. The regression plot of the RNN prediction model further supports this, showing high predictive accuracy with a test set RMSE of 0.0007 and R² of 0.9129, and near-perfect performance on the entire dataset with an RMSE of 0.0005 and R² of 0.9999. These results confirm both the material's wear resistance and the model's reliability in forecasting its coefficient of friction trends.

In contrast to the COOH-MWCNT Epoxy Composite, Fig. 9, the NH₂-MWCNT Epoxy composite exhibits significant periodic fluctuations in the Coefficient of Friction. The COF starts at approximately 0.35–0.40 but undergoes strong, slow oscillations, indicating an adhesive-dominant wear mechanism. The enhanced interfacial bonding from the amine functionalization of the MWCNTs promotes strong adhesion with the counter surface. This leads to a continuous, cyclical process of material transfer and tribofilm formation. The growth and compaction of this film gradually alter the contact conditions, causing the COF to increase. Once the film is worn away or redistributed, the COF decreases, and the cycle repeats. This continuous process of build-up and break-down is responsible for the slow, large-scale fluctuations observed in the COF data. These observations are in good agreement with the SEM analysis of the wear surfaces.

The NH₂-functionalized MWCNT epoxy composite exhibits a pronounced ductile behavior [41], which can lead to severe adhesive wear during sliding. This occurs as material from the composite adheres to the counter ball, forming a protective transfer layer between the ball and the test surface. Over time, this layer breaks down, exposing the ball surface to direct contact, resulting in further wear and smearing of the composite material, as observed in the SEM top-view image. The cross-sectional view of the wear track shows hills and valleys formed by material removal and deformation. This uneven wear morphology causes significant fluctuations in the coefficient of friction over time. Despite this complexity, the predictive RNN model demonstrates a strong fit to the experimental data, with a test set RMSE of 0.0103 and R² of 0.9553, and even better performance on the entire dataset, achieving an RMSE of 0.0077 and R² of 0.9814. These results indicate the model's effectiveness in capturing the dynamic tribological behavior of the NH₂-MWCNT epoxy composite.

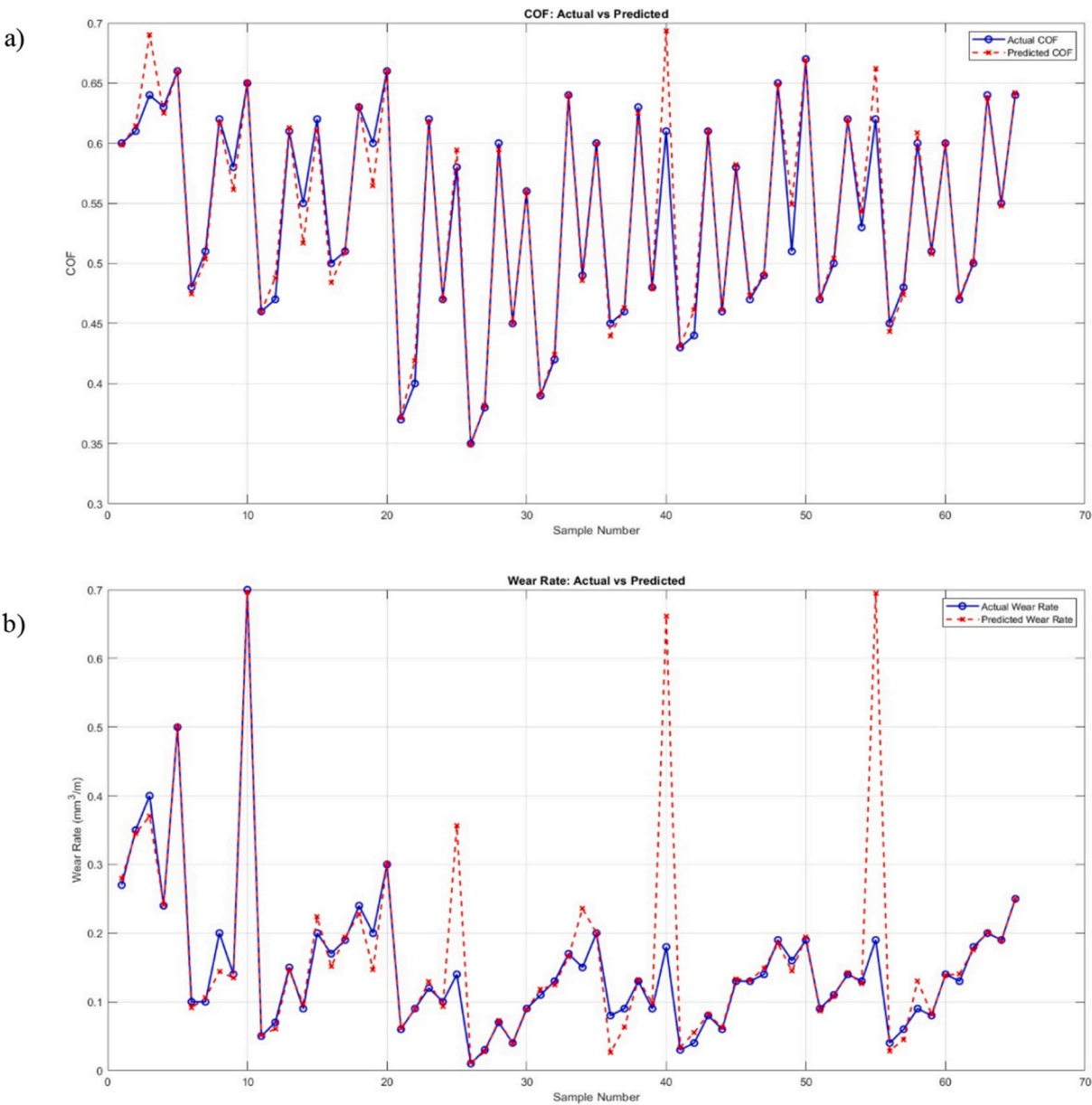


Fig. 5. ANN actual vs Predicted results by sample number a) COF; b) Wear rate.

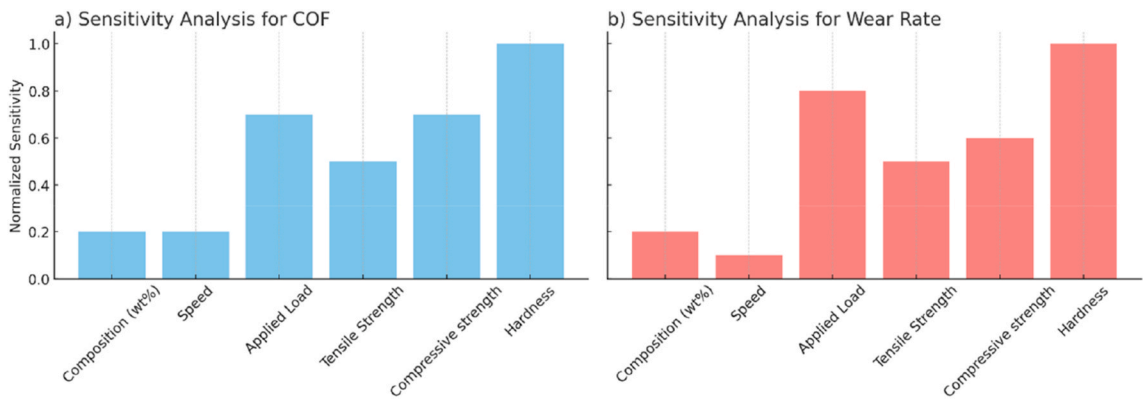


Fig. 6. Sensitivity Analysis of Input Parameters on Tribological Properties a) COF; b) Wear rate.

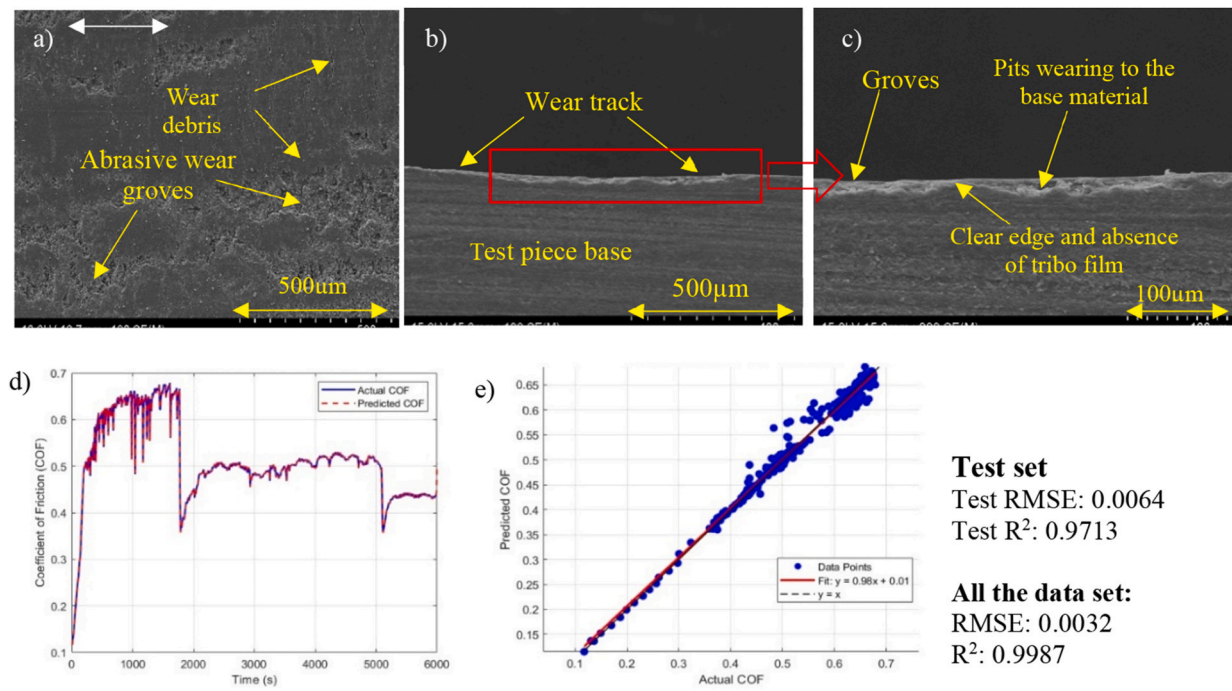


Fig. 7. Wear track top view of P-MWCNTs 6000 S at 10 N, 5 Hz, 400 m test; b-c) Wear track cross-sections at varying magnifications; d) RNN actual vs. predicted COF; e) Regression plot of actual vs. predicted COF over time.

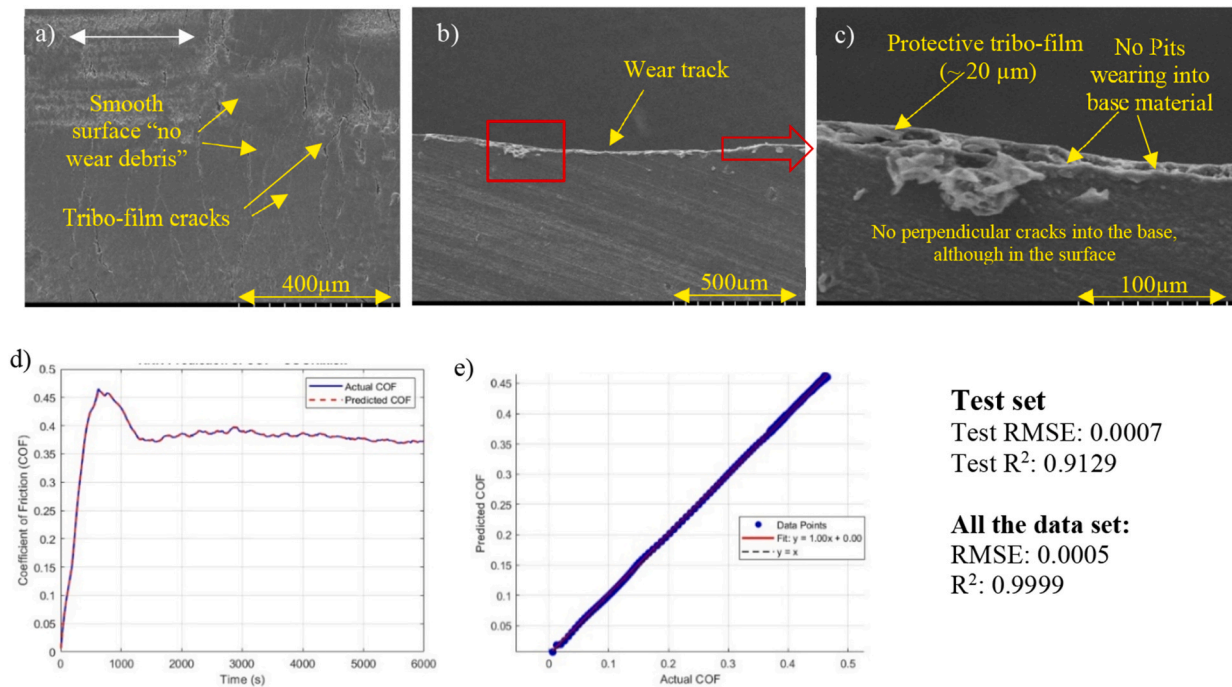


Fig. 8. Wear track top view of COOH-MWCNTs 6000 S at 10 N, 5 Hz, 400 m test; b-c) Wear track cross-sections at varying magnifications; d) RNN actual vs. predicted COF; e) Regression plot of actual vs. predicted COF over time.

COF prediction for the Silane-Functionalized MWCNT-Epoxy Composite (Fig. 10), the Silane-functionalized MWCNT composite exhibited an initial COF rise, peaking at 0.38, followed by a gradual rise to 0.58. The predicted COF aligns well with the experimental data, demonstrating high model accuracy with minimal deviations. This behavior suggests that Silane treatment improves interfacial adhesion between the MWCNTs and epoxy matrix, promoting uniform load transfer and reducing fluctuations. SEM analysis revealed that the wear mechanism is

a combination of severe adhesive wear and material pull-outs. The cross-sectional SEM images show large chunks of wear debris detaching from the surface, indicating significant material removal during the sliding process. This progressive detachment suggests an unstable tribofilm formation, which contributes to fluctuations and gradual increases in friction over time. Despite these complex wear dynamics, the recurrent neural network (RNN) model achieved good predictive accuracy, with a test set RMSE of 0.0101 and an R^2 value of 0.9013. When evaluated on

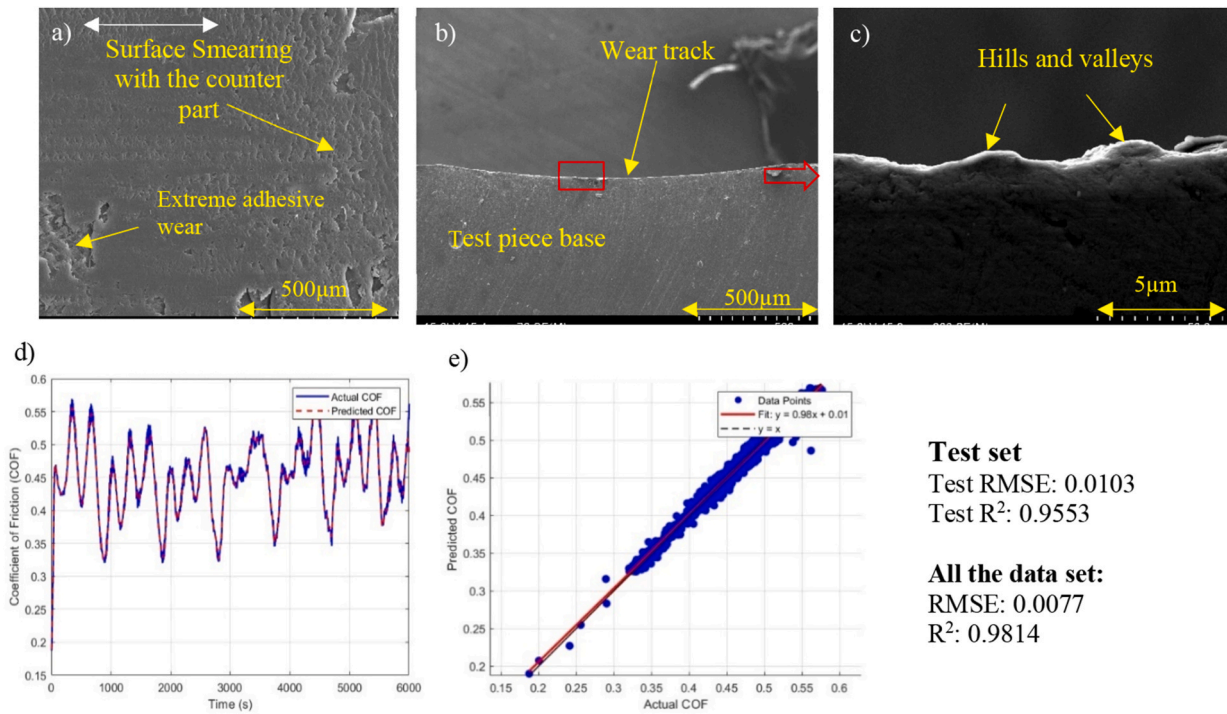


Fig. 9. Wear track top view of NH₂-MWCNTs 6000 S at 10 N, 5 Hz, 400 m test; b-c) Wear track cross-sections at varying magnifications; d) RNN actual vs. predicted COF; e) Regression plot of actual vs. predicted COF over time.

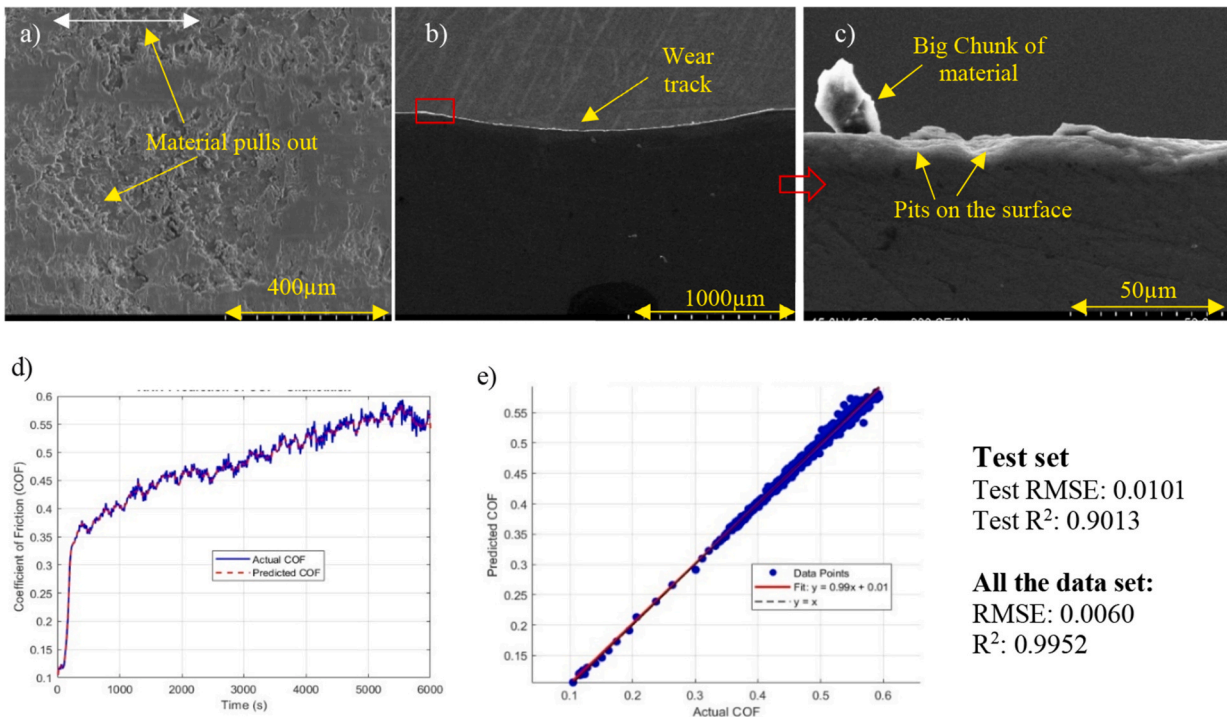


Fig. 10. Wear track top view of Silane modified-MWCNTs 6000 S at 10 N, 5 Hz, 400 m test; b-c) Wear track cross-sections at varying magnifications; d) RNN actual vs. predicted COF; e) Regression plot of actual vs. predicted COF.

the entire dataset, the model's performance improved further, showing an RMSE of 0.0060 and an R^2 of 0.9952. These results demonstrate the model's capability to effectively capture and predict the tribological behavior under unstable tribofilm conditions.

Fig. 11 shows the ANN's COF prediction over 6000 s for neat epoxy, with actual and predicted values fluctuating between 0.61 and 0.7, with

slight fluctuation. The SEM images reveal severe abrasive wear accompanied by consistently high COF throughout the test duration. This high COF is associated with surface layer separation characterized by delamination and material pull-off. Notably, the degree of abrasion is significantly greater in the neat epoxy compared to the P-MWCNT functionalized epoxy composite. The top-view SEM images of the wear

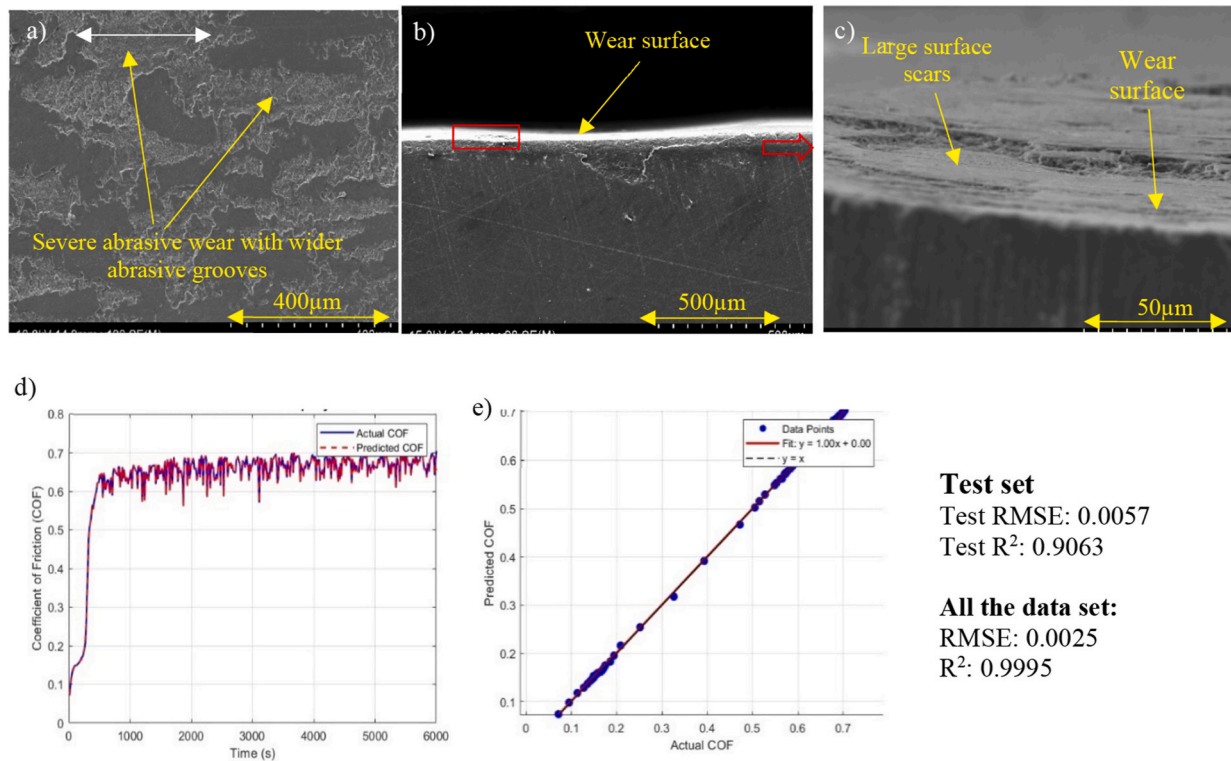


Fig. 11. Wear track top view of neat epoxy 6000 S at 10 N, 5 Hz, 400 m test; b-c) Wear track cross-sections at varying magnifications; d) RNN actual vs. predicted COF; e) Regression plot of actual vs. predicted COF over time.

track display wide abrasive grooves, while cross-sectional SEM images show large wear scars on the wear surface, indicating extensive material removal. The predictive RNN model demonstrated strong performance in capturing this wear behavior, with a test set RMSE of 0.0057 and an R^2 of 0.9063. Evaluation on the entire dataset showed even better accuracy, with an RMSE of 0.0025 and an R^2 of 0.9995, confirming the model's reliability in forecasting friction and wear trends under severe abrasive conditions.

4.5. Convolution neural network

This study evaluates the effectiveness of a CNN model in classifying different wear mechanisms based on SEM image analysis. The CNN was trained to distinguish four distinct wear types: Abrasive Wear, Adhesive Wear, Fatigue Wear, and Material Pull-Outs. Fig. 12a presents key performance metrics, including a confusion matrix, training progress, and loss reduction curves, which demonstrate the model's ability to accurately learn and classify wear patterns. The confusion matrix assesses CNN's classification performance by comparing actual wear classes (true labels) against predicted categories. The matrix shows perfect classification performance, achieving 85 % accuracy for all four wear mechanisms. Such high classification accuracy suggests that the CNN effectively extracted relevant wear features, likely capturing surface texture, morphology, or intensity variations from tribological images. However, perfect classification results can also indicate overfitting, where the model may have learned dataset-specific patterns rather than generalized features. Further testing on external datasets with different wear conditions would be necessary to ensure robustness. Fig. 12b shows the random prediction of 24 data samples by the model.

The training progress Fig. 13 graph illustrates how CNN's classification accuracy improved over multiple iterations. Initially, the training accuracy fluctuated due to random weight initialization and early-stage learning adjustments. However, as training progresses, the model quickly learns relevant features, achieving high accuracy (85 %) of

approximately 145 iterations. The validation accuracy follows a similar trend, confirming that the model generalizes well to the validation dataset without significant overfitting. The loss reduction curve at the bottom provides further insight into the model's learning behavior. Initially, the loss is high, indicating a high prediction error. However, as the training progresses, the loss declines rapidly within the first 50 iterations and approaches near-zero values after approximately 200 iterations. This suggests that the model effectively minimizes classification errors and converges efficiently. The fact that both training and validation loss decrease in parallel indicates that CNN maintains stability and does not suffer from severe overfitting, despite achieving 85 % accuracy.

5. Conclusions

This study leverages Artificial Neural Network (ANN), Recurrent Neural Network (RNN) with Long Short-Term Memory (LSTM) layers, and Convolutional Neural Network (CNN) to predict and classify tribological behaviors of self-lubricating epoxy- Multiwall Carbon nanotube (MWCNT) composites, offering significant implications for mechanical engineering and preventive maintenance. These models enable precise forecasting of friction and wear, critical for optimizing component durability and scheduling maintenance to prevent failures in machinery such as bearings, gears, and seals. By predicting tribological performance, these models reduce downtime, enhance system reliability, and lower maintenance costs in industries reliant on high-performance composites.

The ANN demonstrated robust prediction of Coefficient of Friction (COF) and Wear Rate, with high training ($R = 0.99945$) and validation ($R = 0.97894$) correlations, though testing ($R = 0.72464$) suggested overfitting risks. COF predictions were highly accurate ($R^2 = 0.982385$, RMSE = 0.016603), while Wear Rate showed moderate precision ($R^2 = 0.780183$, RMSE = 0.092527) due to variability. Sensitivity analysis identified hardness as the primary driver for both outputs, guiding

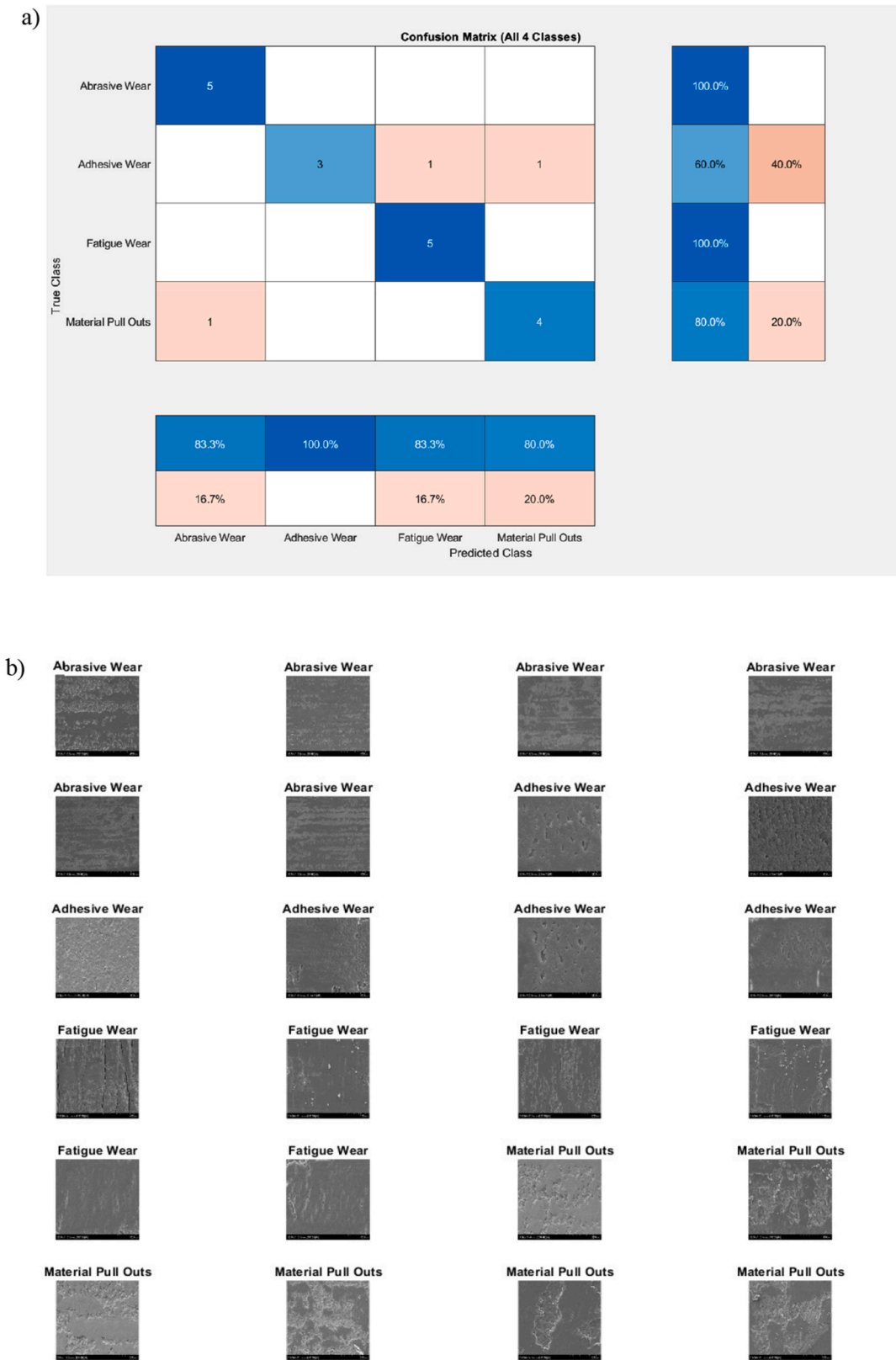


Fig. 12. Convolutional Neural Network Performance in Classifying Wear Mechanisms, a) Confusion Matrix and b) SEM Image Validation.

material design for enhanced wear resistance.

The RNN- Bidirectional LSTM model excelled in capturing time-dependent COF evolution over 6000 s across functionalized MWCNT composites. Epoxy composites with functionalized MWCNTs under 10N load, 5 Hz, and 400m sliding distance, showing various COF and wear mechanisms. For pristine MWCNT, COF fluctuated 0.35–0.7 linked to abrasive wear. COOH-MWCNT showed stable COF around 0.37, indicating strong self-lubrication and protective tribo-film. NH₂-MWCNT exhibited oscillations from adhesive wear. Silane-MWCNT COF rose from 0.38 to 0.58 with unstable tribofilm. Neat epoxy showed high COF

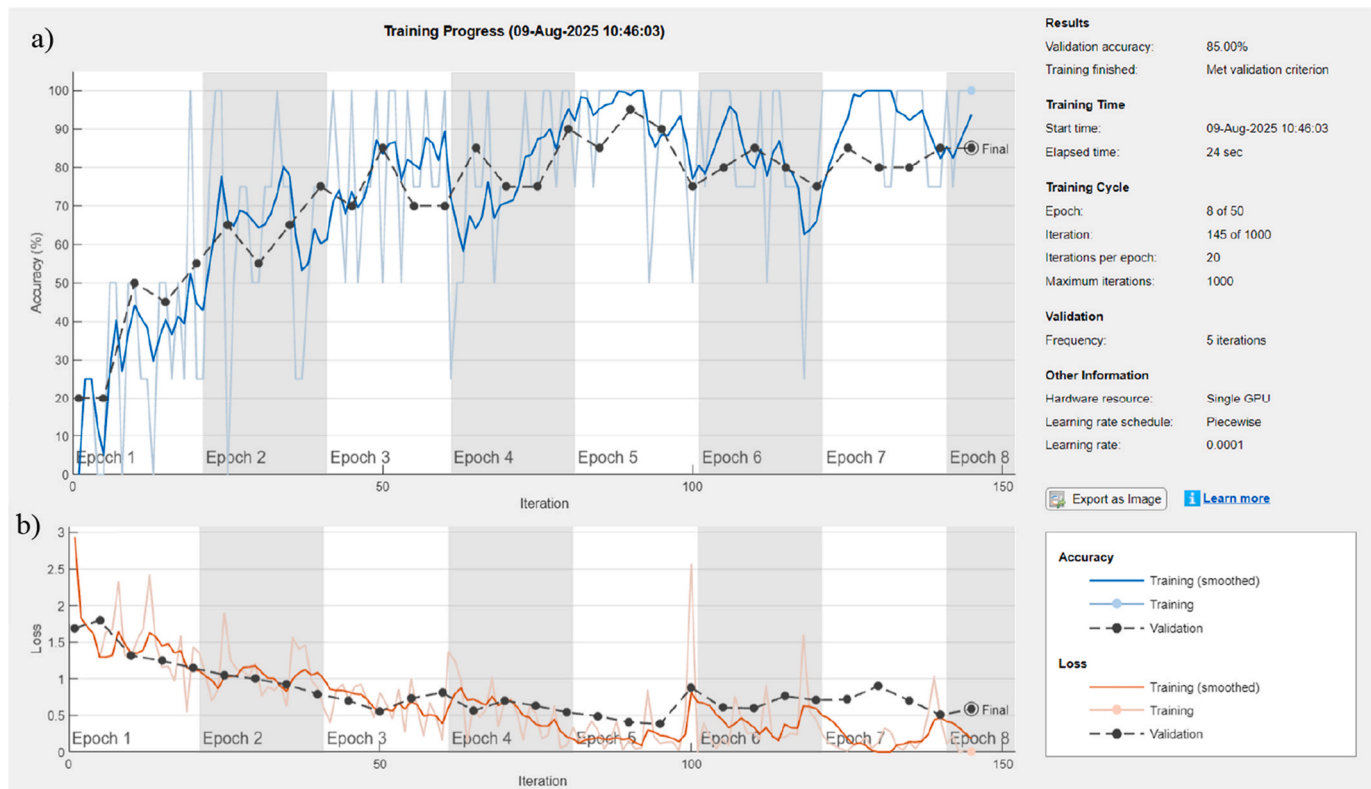


Fig. 13. Convolutional Neural Network Training Progress, a) Accuracy and b) Loss Evolution.

0.61–0.7 and severe abrasion. Overall, the RNN accurately predicted complex tribological behaviors. The best RNN- Bidirectional LSTM model achieve the near-perfect performance on the RMSE of 0.0005 and R^2 of 0.9999.

The CNN achieved 85 % accuracy in classifying abrasive, adhesive, fatigue, and material pull-out mechanisms from 100 of SEM images, though perfect classification suggests potential overfitting, necessitating further validation. These models collectively advance tribological analysis by enabling early detection of wear patterns, optimizing composite formulations, and informing maintenance schedules. Future work should focus on practical applications by aligning test conditions with real-world scenarios. Emphasis should be placed on developing effective preventive maintenance strategies to enhance system reliability and performance, ensuring that research outcomes are directly applicable and beneficial in operational environments.

CRedit authorship contribution statement

Ravisrini Jayasinghe: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Maziar Ramezani:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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